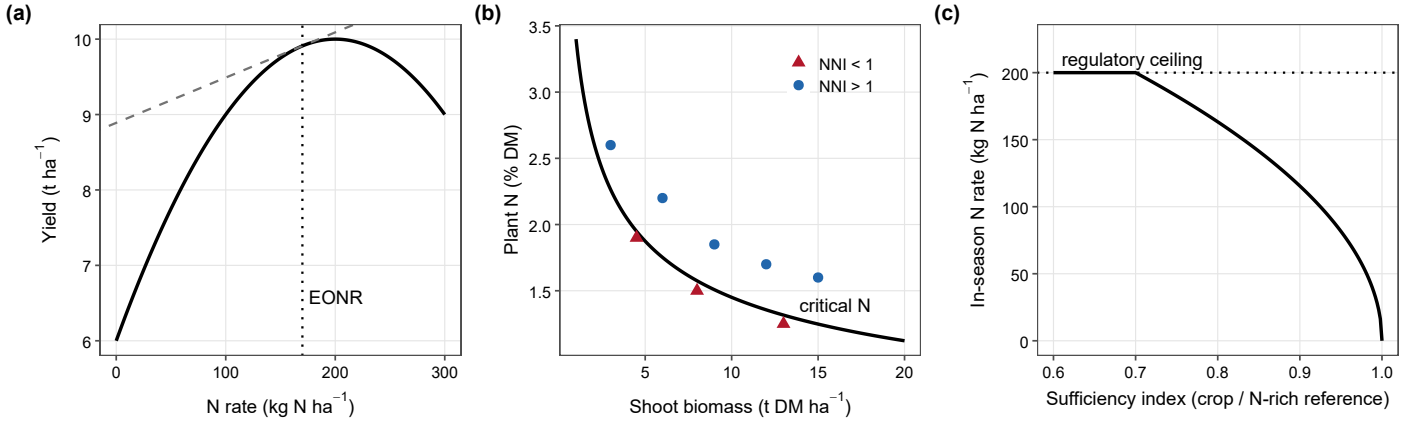
(a)
Yield (t ha^{-1})
10
9
8
7
6
EONR
0
100
200
300
N rate (kg N ha^{-1})
(b)
Plant N (% DM)
3.5
3.0
2.5
2.0
1.5
NNI < 1
NNI > 1
critical N
5
10
15
20
Shoot biomass (t DM ha^{-1})
(c)
In-season N rate (kg N ha^{-1})
regulatory ceiling
200
150
100
50
0
0.6
0.7
0.8
0.9
1.0
Sufficiency index (crop / N-rich reference)